

Triangle Congruence Postulates and CPCTC

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Congruent triangles are triangles that have exactly the same size and shape.

SSS Congruence Postulate

If the three sides of one triangle are congruent to the three corresponding sides of another triangle, then the triangles are congruent.

$$\triangle ABC \cong \triangle DEF$$

if $AB = DE$, $BC = EF$, and $AC = DF$.

SAS Congruence Postulate

If two sides and the **included angle** of one triangle are congruent to the corresponding parts of another triangle, then the triangles are congruent.

The included angle is the angle formed *between* the two known sides.

ASA Congruence Postulate

If two angles and the **included side** of one triangle are congruent to the corresponding parts of another triangle, then the triangles are congruent.

The included side is the side *between* the two known angles.

AAS Congruence Theorem

If two angles and a **non-included side** of one triangle are congruent to the corresponding parts of another triangle, then the triangles are congruent.

Here, the known side is *not* between the two known angles.

HL Congruence Theorem

The **HL Theorem** applies only to **right triangles**.

If the hypotenuse and one leg of a right triangle are congruent to those of another right triangle, then the triangles are congruent.

Congruence Shortcuts at a Glance

SSS — 3 sides

SAS — 2 sides, included angle

ASA — 2 angles, included side

AAS — 2 angles, non-included side

HL — right triangles only

CPCTC

CPCTC means **C**orresponding **P**arts of **C**ongruent **T**riangles are **C**ongruent.

After proving two triangles congruent, *all* corresponding sides and angles are also congruent — not just the parts used in the postulate.

Examples

Example 1

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, and $AC = DF$.

Answer: All three pairs of corresponding sides are congruent, so the triangles are congruent by **SSS**.

Example 2

In $\triangle PQR$ and $\triangle XYZ$, $PQ = XY$, $\angle Q = \angle Y$, and $QR = YZ$.

Answer: $\angle Q$ is between sides PQ and QR , so the triangles are congruent by **SAS**.

Example 3

Given $\triangle ABC \cong \triangle DEF$.

Answer: By **CPCTC**, $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$, and $AB = DE$, $BC = EF$, $AC = DF$.

Practice Exercises

Practice 1

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, and $AC = DF$. Which congruence postulate proves the triangles congruent?

Practice 2

In $\triangle PQR$ and $\triangle XYZ$, $PQ = XY$, $\angle P = \angle X$, and $PR = XZ$. Identify the congruence postulate.

Practice 3

In $\triangle MNO$ and $\triangle RST$, $\angle M = \angle R$, $MN = RS$, and $\angle N = \angle S$. What congruence postulate applies?

Practice 4

In $\triangle GHI$ and $\triangle JKL$, $\angle G = \angle J$, $\angle I = \angle L$, and $HI = KL$. Which theorem proves the triangles congruent?

Practice 5

Two right triangles have hypotenuses measuring 13 cm and one corresponding leg measuring 5 cm. Which theorem proves they are congruent?

Practice 6

Given $\triangle ABC \cong \triangle DEF$, identify the congruent angle corresponding to $\angle B$.

Practice 7

Given $\triangle PQR \cong \triangle XYZ$, identify the side corresponding to QR .

Activities

Activity 1

In $\triangle UVW$ and $\triangle XYZ$, $UV = XY$, $VW = YZ$, and $UW = XZ$. Determine the congruence postulate.

Activity 2

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $\angle B = \angle E$, and $BC = EF$. Identify the congruence postulate.

Activity 3

In $\triangle JKL$ and $\triangle MNP$, $\angle J = \angle M$, $JK = MN$, and $\angle K = \angle N$. Determine the congruence postulate.

Activity 4

In $\triangle QRS$ and $\triangle TUV$, $\angle Q = \angle T$, $\angle R = \angle U$, and $QS = TV$. Identify the congruence theorem.

Activity 5

Two right triangles have hypotenuses measuring 17 cm and one corresponding leg measuring 8 cm. Which theorem proves they are congruent?

Activity 6

Given $\triangle GHI \cong \triangle JKL$, identify the congruent angle corresponding to $\angle H$.

Activity 7

Given $\triangle ABC \cong \triangle PQR$, identify the side corresponding to AC .

Thank you!